63abb7fe-c73a-442a-b515-f0ce0f7c0f70

<current>

<request>

You are working on an issue in the 'musicmonk42/The_Code_Factory_Working_V2' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/The_Code_Factory_Working_V2/The_Code_Factory_Working_V2 (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-02-15T01:54:40.710Z</current_datetime>

Consider the following problem statement:

<problem_statement>
Problem Statement

The application is experiencing multiple critical failures preventing proper operation:

1. Audit Crypto KMS Key Mismatch (CRITICAL)

Symptoms:
- `InvalidCiphertextException: Master key encrypted with different KMS key`
- Falls back to `DummyCryptoProvider` (no real security)
- Audit log integrity compromised
- Repeated error: `raise _SOFTWARE_KEY_MASTER_LAST_ERROR`

Root Cause:
The `AUDIT_CRYPTO_SOFTWARE_KEY_MASTER_ENCRYPTION_KEY_B64` environment variable contains a ciphertext that was encrypted with a different AWS KMS key than the one currently configured (`alias/audit-log-key`). The system cannot decrypt this value.

Expected Behavior:

````
For Railway deployments (no AWS KMS), the system should:
- Detect that `USE_ENV_SECRETS=true` is set
- Use the base64-encoded **plaintext** master key directly from
the environment variable
- NOT attempt KMS decryption for Railway deployments

## 2. Kafka Bridge Initialization Failure

**Error:**
```
Failed to start Kafka bridge: AIOKafkaProducer.__init__() got an
unexpected keyword argument 'retries'
```

**Root Cause:**
The `aiokafka` library version doesn't support the `retries`
parameter that's being passed during initialization.

**Impact:**
- Message bus falls back to local queue only
- Distributed message handling disabled
- Jobs may vanish on restart

## 3. Production Package/Data Gaps

**Errors:**
- `No module named 'runner.language_utils'` → Clarifier falls
back to rule-based
- `vader_lexicon not found` → Sentiment analysis features
degraded

**Impact:**
- Reduced ML/NLP functionality
- Degraded user experience

---

## Required Changes

### File: `generator/audit_log/audit_crypto/secrets.py`

**Fix the secret manager to detect Railway environment and skip
KMS decryption:**

```python
class EnvVarSecretManager(SecretManager):
    """
````
````

```
    Secret manager that reads from environment variables.

    For Railway/PaaS deployments:
    - Detects USE_ENV_SECRETS=true
    - Returns plaintext secrets directly (Railway encrypts at
platform level)
    - Skips AWS KMS decryption

    For AWS deployments:
    - Attempts KMS decryption if AWS credentials are available
    """

    async def get_secret(self, secret_name: str) ->
Optional[str]:
        value = os.getenv(secret_name)
        if not value:
            return None

        # Railway/PaaS mode: return plaintext directly
        if os.getenv("USE_ENV_SECRETS", "").lower() == "true":
            logger.info(f"EnvVarSecretManager: Using Railway/
PaaS mode for {secret_name}")
            return value

        # AWS mode: attempt KMS decryption
        if secret_name ==
"AUDIT_CRYPTO_SOFTWARE_KEY_MASTER_ENCRYPTION_KEY_B64":
            try:
                # Check if AWS credentials are available
                if not (os.getenv("AWS_ACCESS_KEY_ID") or
os.getenv("AWS_REGION")):
                    logger.info(f"No AWS credentials found,
treating {secret_name} as plaintext")
                    return value

                # Attempt KMS decryption
                import boto3
                kms = boto3.client('kms',
region_name=os.getenv("AWS_REGION", "us-east-1"))
                response =
kms.decrypt(CiphertextBlob=base64.b64decode(value))
                return
base64.b64encode(response['Plaintext']).decode()
            except Exception as e:
                logger.warning(f"KMS decryption failed for
{secret_name}, using as plaintext: {e}")
                return value
```

````
        return value
```

### File: `omnicore_engine/message_bus/kafka_bridge.py`

**Remove the incompatible `retries` parameter from
AIOKafkaProducer initialization:**

Search for:
```python
self.producer = AIOKafkaProducer(
    bootstrap_servers=self.config.kafka_bootstrap_servers,
    retries=5,  # <-- REMOVE THIS LINE
    ...
)
```

Replace with:
```python
self.producer = AIOKafkaProducer(
    bootstrap_servers=self.config.kafka_bootstrap_servers,
    # Note: retries removed - not supported in current aiokafka
version
    # Retry logic should be handled at application level
    ...
)
```

### File: `Dockerfile`

**Add missing NLTK data download:**

Add before the runtime stage:
```dockerfile
RUN python -m nltk.downloader -d /opt/nltk_data vader_lexicon
punkt stopwords
```

### File: `generator/audit_log/audit_crypto/
audit_crypto_factory.py`

**Improve error handling and logging for KMS failures:**

Around line 593 in `_ensure_software_key_master()`, update the
error message to be more helpful:
````
````

```python
except Exception as e:
    error_msg = f"Failed to decrypt software key master: {e}"

    # Provide specific guidance based on error type
    if "InvalidCiphertextException" in str(e):
        if os.getenv("USE_ENV_SECRETS", "").lower() == "true":
            error_msg += "\n\nRailway/PaaS Deployment Detected:"
            error_msg += "\n1. Ensure
AUDIT_CRYPTO_SOFTWARE_KEY_MASTER_ENCRYPTION_KEY_B64 contains a
base64-encoded PLAINTEXT key"
            error_msg += "\n2. Generate a new key: python -c
\"import base64, os;
print(base64.b64encode(os.urandom(32)).decode())\""
            error_msg += "\n3. Set the generated key directly in
Railway environment variables"
            error_msg += "\n4. DO NOT encrypt the key with KMS
for Railway deployments"
        else:
            error_msg += "\n\nAWS Deployment:"
            error_msg += "\n1. The encrypted key was created
with a different KMS key"
            error_msg += "\n2. Generate a new key and encrypt
with current KMS key"
            error_msg += f"\n3. Current KMS key:
{os.getenv('AUDIT_CRYPTO_KMS_KEY_ID', 'NOT SET')}"

    logger.error(error_msg)
    raise CryptoInitializationError(error_msg) from e
```

File: `railway.toml`

Add Railway-specific environment variables:

```toml
[deploy]
# Railway-specific secret management
USE_ENV_SECRETS = "true"

# Clarify that KMS is not used on Railway
AUDIT_CRYPTO_USE_KMS = "false"
```

Testing Steps

```
After these changes:

1. **Railway Deployment:**
   - Generate a new plaintext base64 key: `python -c "import
base64, os; print(base64.b64encode(os.urandom(32)).decode())"`
   - Set `AUDIT_CRYPTO_SOFTWARE_KEY_MASTER_ENCRYPTION_KEY_B64`
to this plaintext key in Railway
   - Set `USE_ENV_SECRETS=true` in Railway
   - Redeploy and verify no KMS errors

2. **Kafka:**
   - Verify logs show: `Kafka bridge initialized successfully`
   - NOT: `Failed to start Kafka bridge: ...retries...`

3. **ML Features:**
   - Verify no errors about missing `vader_lexicon`
   - Verify Clarifier uses proper sentiment analysis

---

## Success Criteria

- ✅ No `InvalidCiphertextException` errors
- ✅ Audit crypto uses `SoftwareCryptoProvider` (not Dummy)
- ✅ Kafka bridge initializes successfully
- ✅ NLTK data available in production
- ✅ Jobs complete without "job not found" errors
- ✅ Logs show: `Cryptographic configuration validated
successfully`

---

## Priority: CRITICAL

This issue prevents:
- Proper audit logging (security compliance)
- Distributed job processing (Kafka)
- Full ML/NLP functionality
- Production deployment reliability
</problem_statement>
```

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.

A **CI/Build Failure** has been mentioned in the problem statement, make sure to follow your instructions for addressing CI/Build Failures.